

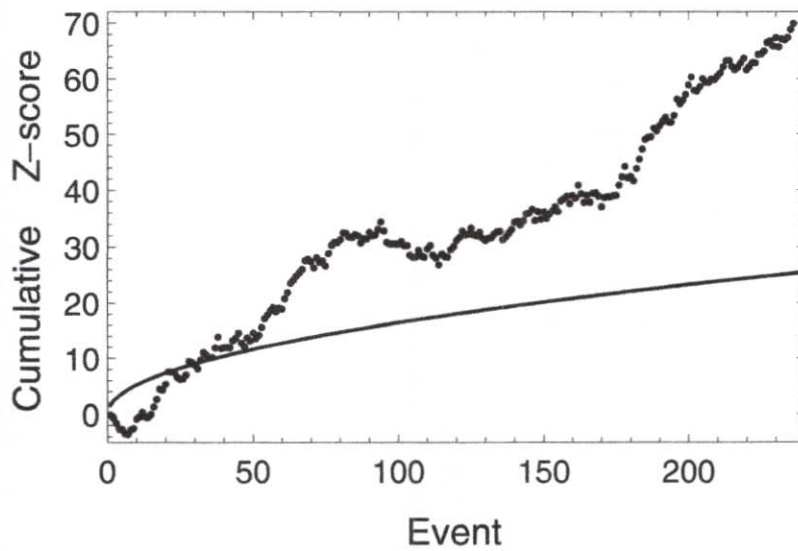


Fig. 3. The summed chronological deviation of 236 event z-scores. Each point represents the contribution from the normal z-score of a single event. The null expectation is zero and the parabola gives a 5% probability envelope for positive deviation. The relatively steady trend shows that the cumulative deviation is broadly distributed among many events and is not due to outlier events.

visualizations of the distribution: the chronological cumulative sum and the sorted cumulative distribution function.

The formal experimental result combines diverse event predictions as equally weighted z-scores, Z_E . As such, it does not provide information about which secondary parameters drive the measured effect. The relative importance of event types, durations and the different deviation statistics can only be addressed by a further analysis of the event data. However, a number of conclusions can be drawn from the distribution of the event z-scores. First, the event effect size of 0.30 is broadly distributed and small. This means that although an individual event is likely to contribute to the aggregate result, there is not sufficient statistical power to meaningfully interpret a single event statistic and analyses must therefore use large event sets. For example, an idealized model which attributes a uniform population mean of 0.3 to events would require at least 70 events to attain a power of 80% at the 95% confidence level. Second, given that there are no outliers, the small effect size implies that statistical noise will determine the *distribution* of event z-scores. An immediate consequence is that the distribution shape will be approximately Gaussian. Figure 4 shows that, aside from a shifted mean value, the event z-scores do have the form of a normal distribution. We find that the z-scores fit reasonably well to a standard normal distribution with mean 0.3, and that tests of the variance, skewness and kurtosis

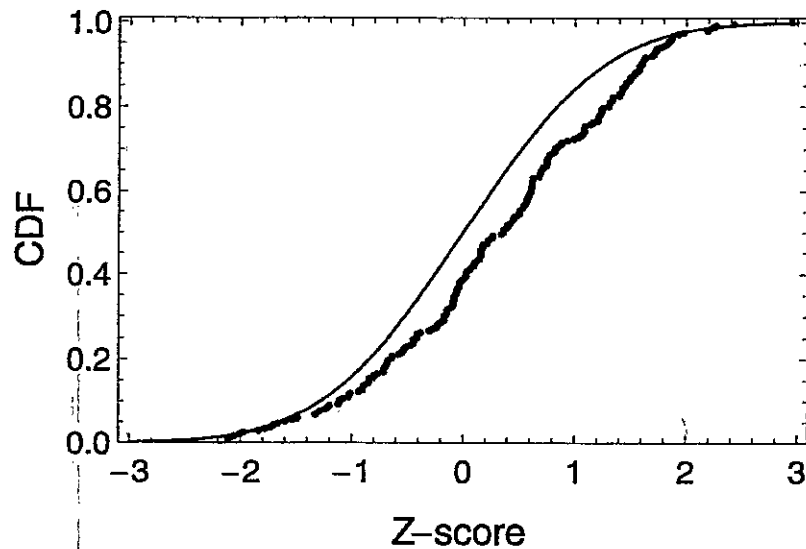


Fig. 4. The plot compares the (sorted) cumulative distribution function of event z-scores with the standard normal distribution. The formal result is evidenced as a positive shift of the event z-scores along the horizontal axis. The median z-score is 0.365, as opposed to zero for the null hypothesis.

also accord with a normal distribution of variance = 1. Third, the mean event z-score represents a *lower bound* of the measured effect. The positive aggregate result, which is in accord with the hypothesis, may nevertheless include some events that produce actual negative deviations. In this case, the magnitude of the true event effect size will be larger than our measured value. It is also plausible, given the exploratory nature of the event selection procedure, that a number of events might correspond to true null data periods.²³

We can develop these conclusions to estimate the fractions of events with null or negative variance deviation via a simple model. As discussed above, the event z-scores distribute normally about the experimental mean. A model for the fractions of positive, negative and null events can be constructed by taking the sum of three standard normal Gaussian distributions with positive, zero and negative means, respectively. The means and optimal weights of the model Gaussians are determined by fitting the model to the distribution of experimental z-scores. Details are described in Appendix 2. The model yields a region of preferred fractional composition which includes potential contributions from both null and negative-going events. We estimate a positive event fraction of 67%, with 16% and 17% for the fractions of null and negative deviation events, respectively. The corresponding means for positive and negative deviation events are 0.56 ± 0.09 and -0.49 ± 0.20 (1-sigma uncertainties). The model thus finds a mean effect size for the positive deviation events that is substantially

larger than the average event z-score of 0.30 ± 0.08 . A conservative conclusion is that the model provides quantitative evidence for the reasonable supposition that the mean event z-score is a lower bound to the event effect size. It also underscores the possibility that the sign of the variance deviations may be negative for a minority fraction of events.

The experimental variance statistics measure different kinds of data deviations and we would like to know which of these drive the aggregate result, Z_{Tot} . The formal experiment selects a variance recipe depending on the registry specifications for each event. Different variance statistics can be compared by applying them uniformly to the events and calculating the resulting Z_{Tot} .

The network variance for different time blockings can be written compactly as a function of the network autocorrelation.²⁴ As expressed in Equation 4.2, the chi-squared variance statistic for an event is the sum of the squared block z-scores:

$$\chi^2 = \sum_B Z_B^2 \quad (4.4)$$

For network blockings of length T , this can be decomposed into a term proportional to the 1-second network variance and a term γ which is a function of the autocorrelation.

$$\chi^2 = \frac{\chi_0^2}{T} + \frac{\gamma_T}{T} \quad (4.5)$$

The chi-squared variance is converted to a z-score for the event as:

$$Z_E = \frac{Z_0}{\sqrt{T}} + \frac{1}{\sqrt{2T^3}} \sum_l (T-l)\rho(l) \quad (4.6)$$

Here, Z_0 is the event z-score at 1-second blocking and $\rho(l)$ is the autocorrelation of the 1-second z-scores $Z_{B=1}$ for lag l . The decomposition shows that, for zero autocorrelation, the second term is identically zero and the measured event z-score decreases as the inverse square root of the block length. Figure 5 shows that the aggregate Z_{Tot} for the network variance falls within a 1-sigma envelope of the expected theoretical decrease for the case $\rho = 0$, demonstrating that temporal network autocorrelations are not present in the network-blocked event data. This calculation uses 212 events, instead of the full 236. We remove 19 New Year's events and 5 events which have extremely long durations and disproportionate weight, to allow uniform application of the variance recipes.

Results for the device variance for a selection of blockings, T , are shown in Figure 6. The device variance has been calculated explicitly for unique blocking choices because trial vacancies (nulls) complicate a more general treatment.

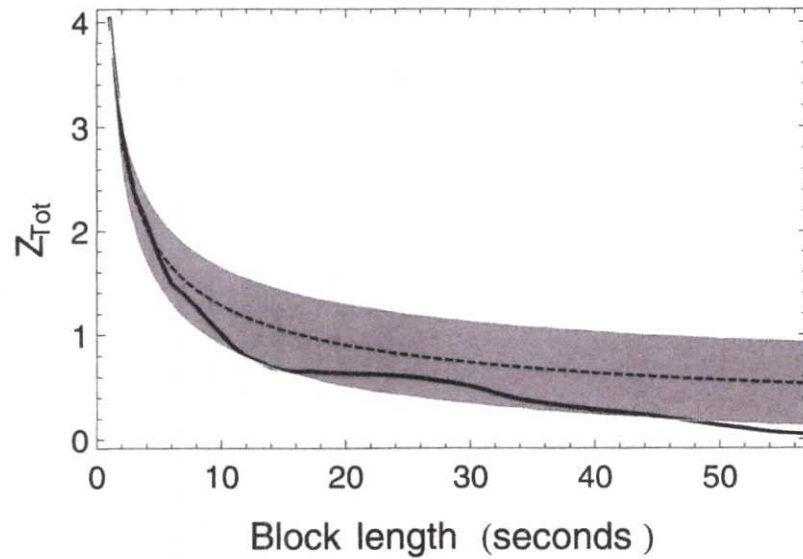


Fig. 5. The z-score, Z_{Tot} for 212 events using network variance recipes. The aggregate z-score is calculated for time blockings ranging from 1 second to 1 minute. The value of Z_{Tot} is largest for 1-second blocking and remains within a 1-sigma envelope of theoretical expectation assuming zero autocorrelation. The plot indicates that the spatial network correlations do not persist beyond the 1-second time-scale.

(Trial vacancies do not affect the network blocking, which produces non-null data seconds for all events.) The blockings in this illustration extend from 1 second to the full duration of the event. In all cases, the magnitude of the mean event z-score is less than 0.1, which is within 1.7 standard deviations of expectation. This is far below the formal result, which has a mean z-score of 0.30. The different device variance estimates are not independent; the longer times include all shorter blocks. The dependence is strongest when the block size ratios are small, as is seen for the similar blocking results in the range of 120 to 900 seconds. We conclude that the device variance statistic (or equivalently, RNG autocorrelation) does not deviate significantly over the event data.

As seen in Figure 6, the trial variance, with blocking ($R = 1$, $T = 1$), is within a 1-sigma deviation from expectation. It is instructive to examine this minimal blocking more closely. Trial blocking assigns a single trial to each block and is insensitive to correlations between trials. Trial-blocked statistics are thus descriptive of the individual RNG behavior (they give the RNG state probabilities) and provide information on the RNG outputs exclusive of any correlation. Table 2 lists standard descriptive statistics of the event trials. All are consistent with null expectation, which indicates that the outputs of the RNG devices *do not deviate from standard normality over the events*. In contrast to the

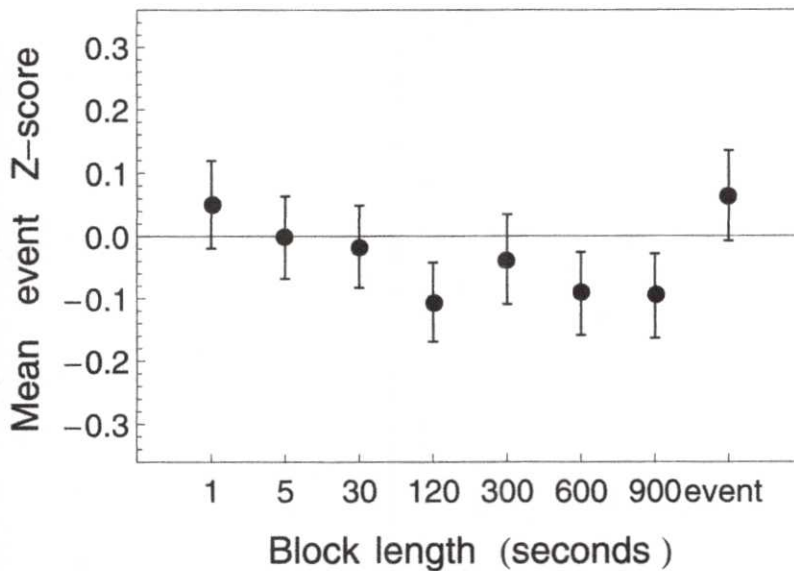


Fig. 6. Average device variance z-scores for events, calculated for various block lengths ranging over 1 second to 15 minutes, and for the whole event.

intending-subject experiments, we find no anomalous deviations in the individual RNG behavior.

These analyses demonstrate that the formal result is driven only by the 1-second network variance, while the RNG state probabilities and autocorrelation conform to expectation. The network variance can be decomposed to show its relation to synchronized RNG-RNG correlations. The chi-squared variance, in terms of the RNG trial z-scores, $z_{r,t}$, is the sum of the trial variance, $Var[z]$, and a summation of trial pair-products.

$$\chi^2 = \sum_t \left(\frac{1}{\sqrt{N}} \sum_r z_{r,t} \right)^2 \quad (4.7)$$

$$= \frac{1}{N} \sum_t \sum_{i,j} z_{i,t} z_{j,t} \quad (4.8)$$

$$= \frac{1}{N} \sum_{i \neq j} \left(\sum_t z_{i,t} z_{j,t} \right) + T_0 Var[z] \quad (4.9)$$

Here, the (i,j) label RNGs and T_0 is the length of the event, in seconds. Then,

$$\approx (N-1)T_0 \overline{\langle z_i z_j \rangle} + T_0 Var[z] \quad (4.10)$$

where the overstrike is an average over all seconds, T_0 , and the brackets indicate an average over unique pairs of RNGs, which yields

TABLE 2
Descriptive Statistics for the Event Data RNG Trials

Statistic	Event weighting			Trial weighting		
	Value	Expect	Z	Value	Expect	Z
Mean	0.00014	0	0.69	0.00006	0	0.98
Variance	1.00045	1	0.95	0.999998	1	-0.02
Skewness	0.00083	0	0.51	0.000103	0	0.69
Kurtosis	2.9908	2.99	0.11	2.99006	2.99	0.19

The table lists the measured value, theoretical expectation and deviation from expectation (as a normal z-score) for trial statistics based on the first four distribution moments. The event-weighted values give the trial statistics' mean over all the 212 events. That is, the statistic is calculated separately for each event and the resultant values are averaged. Trial-weighted values are calculated for all 274 million trials in the event data combined as a single data set.

$$\approx NT_0(\langle r_{ij} \rangle + \text{Var}[z]) \quad (4.11)$$

where we identify the time average of pair-products with the Pearson correlation, r_{ij} for RNGs i and j , and $\langle r_{ij} \rangle$ with the average of the RNG-RNG correlations over all RNG pairs. The pair-product averages can be approximated by the average of Pearson correlations since we have determined that the trial z 's follow normal statistics and the event lengths satisfy $T_0 \gg 1$. Furthermore, deviations in the 1-second network variance must be dominated by the correlation term, since the expected fluctuations of $\text{Var}[z]$ are relatively small, being of order $1/\sqrt{N}$,

$$\delta\chi^2 \approx NT_0\delta\bar{r}_{ij} + T_0\delta\text{Var}[z] \quad (4.12)$$

$$\approx NT_0 \frac{\sqrt{2}}{T_0^{1/2}N} + T_0 \frac{\sqrt{2}}{(T_0N)^{1/2}} \quad (4.13)$$

$$\approx T_0^{1/2}(1 + O(N^{-1/2})) \quad (4.14)$$

and the measured $\text{Var}[z]$ is within a standard deviation of its null expectation. The value of Z_{Tot} for the network variance recipe on the 212-event subset is $Z = 4.10$. This assumes a theoretical zero mean of blocked z-scores (Equation 4.2). The network variance deviation Z_{Tot} can also be determined empirically from a random re-sampling analysis on the entire database. Using this re-sampling-derived expectation yields a value of $Z = 4.29$, which is comparable to the theoretical value. Details of the re-sampling procedure are given in Appendix 3.

We conclude from the blocking analysis that the aggregate formal result, Z_{Tot} , is driven by inter-RNG correlations on a 1-second time-scale. This is the second major result of the experiment. It implies that the network deviations are due to anomalous correlations between RNGs separated, on average, by thousands of kilometers. This is, moreover, a new class of result, which could only be observed in a distributed network of RNGs.